WARFARIN

High-alert drug
Teratogenic

CHEMICAL NAME (RS)-4-hydroxy-3-(3-oxo-1-phenylbutyl)-2H-chromen-2-one (sodium)
CLASS Oral anticoagulant — vitamin K antagonist (coumarin)
TRADE NAMES Coumadin · Marevan · Jantoven · Warfarin sodium

THERAPEUTIC USES / INDICATIONS

- Mechanical heart valves (drug of choice; DOACs contraindicated).
- Atrial fibrillation with rheumatic mitral stenosis; AF in general (alternative to DOACs).
- Treatment and secondary prevention of DVT/PE.
- Antiphospholipid syndrome; LV thrombus.

ADMINISTRATION

- **PO** once daily at the same time each day (usually evening, so dose can be adjusted after INR result).
- Tablets colour-coded by strength (e.g. 1 mg brown, 3 mg blue, 5 mg pink) — check.
- Overlap with heparin/LMWH for ≥5 days and until INR in range for 24 h (VTE).

DOSAGES

- **Induction:** typically 5–10 mg on day 1 (lower in elderly, liver disease, heart failure), then adjust to INR per protocol.
- **Maintenance:** usually 2–10 mg/day guided by INR.
- **Target INR:** 2.5 (2–3) for VTE, AF; 3.0 (2.5–3.5) for mechanical mitral valves / recurrent VTE on warfarin.
- INR monitoring: daily–alternate days at start, then up to every 12 weeks when stable.

PHARMACOKINETICS

- **Onset:** 24–72 h; full effect 5–7 days. **Duration:** 2–5 days.
- Highly protein-bound (99%); metabolised by CYP2C9 (S-isomer), 1A2, 3A4; genetic variation (VKORC1, CYP2C9).
- Excreted in urine as metabolites. **t½:** 36–42 h.

PHARMACODYNAMICS

- Inhibits **vitamin K epoxide reductase (VKORC1)** → ↓ synthesis of functional clotting factors II, VII, IX, X and proteins C and S.
- Early fall in protein C → transient prothrombotic state (need heparin bridging).

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Bruising, minor bleeding (gums, nose).
- Nausea.
- Hair loss, rash.

ADVERSE EFFECTS (SERIOUS)

- Major haemorrhage (GI, intracranial).
- Skin necrosis (protein C deficiency, early treatment).
- Purple toe syndrome; calciphylaxis.
- Teratogenic (warfarin embryopathy).

CONTRAINDICATIONS & CAUTIONS

- Pregnancy (especially weeks 6–12 and near term); switch to LMWH.
- Active bleeding; haemorrhagic stroke; severe uncontrolled hypertension.
- Recent CNS/eye surgery; bleeding disorders.
- Poor adherence or inability to monitor INR.

KEY INTERACTIONS

- **↑ INR:** amiodarone, metronidazole, co-trimoxazole, fluconazole, macrolides, ciprofloxacin, alcohol binge, paracetamol (regular), NSAIDs (bleeding).
- **↓ INR:** rifampicin, carbamazepine, phenytoin, St John's wort, vitamin K-rich foods.
- **Antiplatelets, SSRIs:** ↑ bleeding.
- Any new drug → check INR within 3–5 days.

NURSING MANAGEMENT

- Check INR result before giving each dose; document in anticoagulant chart.
- Over-anticoagulation: follow protocol (hold, vitamin K PO/IV, prothrombin complex concentrate for major bleeding).
- Monitor for bleeding signs; neuro checks after falls.
- Review every new medication for interactions.
- Provide anticoagulant alert card and booklet.

PATIENT EDUCATION

- Take at the same time daily; never double a missed dose.
- Keep vitamin K intake (green leafy vegetables) consistent — do not suddenly change diet.
- Avoid alcohol excess, aspirin/NSAIDs and herbal remedies unless approved.
- Report bleeding, black stools, severe headache; carry your alert card; avoid pregnancy.

KEY POINTS

- Vitamin K antagonist; factors II, VII, IX, X.
- Countless interactions — check INR with any new drug.
- INR target 2–3 (mechanical mitral 2.5–3.5).
- Reverse with vitamin K ± PCC; teratogenic.